Since the vertical and horizontal directions are equivalent in an image, this algorithm can also be carried out by transforming the columns first and then transforming the rows. Regardless of the order used, the result is the same. From the way that the FFT keeps track of the data, the amplitudes of the low frequency components end up being at the corners of the two-dimensional spectrum, while the high frequencies are at the center. The inverse Fourier transform of an image is calculated by taking the inverse FFT of each row, followed by the inverse FFT of each column (or vice versa).

Figure 24-9 shows an example Fourier transform of an image. Figure (a) is the original image, a microscopic view of the input stage of a 741 op amp integrated circuit. Figure (b) shows the real and imaginary parts of the frequency spectrum of this image. Since the frequency domain can contain negative pixel values, the grayscale values of these images are offset such that negative values are dark, zero is gray, and positive values are light. The low-frequency components in an image are normally much larger in amplitude than the high-frequency components. This accounts for the very bright and dark pixels at the four corners of (b). Other than this, the spectra of *typical* images have no discernable order, appearing random. Of course, images can be *contrived* to have any spectrum you desire.

As shown in (c), the polar form of an image spectrum is only slightly easier to understand. The low-frequencies in the magnitude have large positive values (the white corners), while the high-frequencies have small positive values (the black center). The phase looks the same at low and high-frequencies, appearing to run randomly between $-\pi$ and π radians.

Figure (d) shows an alternative way of displaying an image spectrum. Since the spatial domain contains a *discrete* signal, the frequency domain is *periodic*. In other words, the frequency domain arrays are duplicated an infinite number of times to the left, right, top and bottom. For instance, imagine a tile wall, with each tile being the $N \times N$ magnitude shown in (c). Figure (d) is also an $N \times N$ section of this tile wall, but it straddles four tiles; the center of the image being where the four tiles touch. In other words, (c) is the same image as (d), except it has been shifted $N/2$ pixels horizontally (either left or right) and $N/2$ pixels vertically (either up or down) in the periodic frequency spectrum. This brings the bright pixels at the four corners of (c) together in the center of (d).

Figure 24-10 illustrates how the two-dimensional frequency domain is organized (with the low-frequencies placed at the corners). Row $N/2$ and column $N/2$ break the frequency spectrum into four quadrants. For the real part and the magnitude, the upper-right quadrant is a mirror image of the lower-left, while the upper-left is a mirror image of the lower-right. This symmetry also holds for the imaginary part and the phase, except that the values of the mirrored pixels are opposite in sign. In other words, every point in the frequency spectrum has a matching point placed symmetrically on the other side of the center of the image (row $N/2$ and column $N/2$). One of the points is the *positive* frequency, while the other is the matching